



I. Introduction

- The "Data Trap" Problem: Government agencies and planning bodies generate millions of unstructured PDF documents (project reports, surveys, gazettes) annually. These documents contain critical location-specific data (e.g., "Bridge construction progress in Dibrugarh, Assam") that remains "trapped" in text format, effectively invisible to geospatial analysis tools.
- The Manual Bottleneck: Extracting this spatial intelligence currently requires manual reading and data entry, which is prohibitively time-consuming and error-prone. This creates a disconnect between Natural Language (reports) and Spatial Data Infrastructures (maps).
- The Proposed Solution: We present a novel, privacy-focused Integrated Framework that automates the pipeline from "Raw PDF" to "Interactive Map."
- Key Innovations:
 - Fully Offline Execution: Unlike cloud-based APIs, our system runs locally using Open Source LLMs (Mistral-7B), ensuring strict data privacy for sensitive government files.
 - Indian Context Awareness: Specifically designed to handle the hierarchical administrative structure of India (State to District to Tehsil).
 - Democratization: Enables non-technical policymakers to visualize complex textual reports on ISRO Bhuvan maps without requiring GIS expertise.

II. Literature Review

- Geospatial NER: Standard models fail on Indian administrative hierarchies. We fine-tuned a spaCy transformer on Indian Census & LGD data to resolve ambiguities (e.g., distinguishing a District from a Town).
- RAG for PDFs: Standard RAG systems struggle with complex PDF layouts (tables/columns). Our framework uses OCR & structural parsing before vectorization to ensure accurate, context-aware summaries.
- Research Gap: Existing tools are either cloud-dependent or lack spatial-semantic integration. This work bridges NLP and GIS in a fully offline, secure environment.

III. Problem Statement

- The Unstructured Data Silo: Massive volumes of geospatial intelligence are locked inside unstructured PDF narratives, complex multi-column tables, and scanned notices, rendering them invisible to standard GIS software.
- Inefficient Manual Workflows: Extracting location-specific insights (e.g., "Project X delayed in District Y") currently necessitates tedious manual reading and data entry, which is unscalable and highly prone to human error.
- Linguistic & Spatial Ambiguity: Generic global AI models struggle with the Indian context. They often fail to distinguish between identical place names across states (e.g., Aurangabad) or understand administrative hierarchies (Tehsil vs. District).
- Lack of Semantic Context: Merely plotting a coordinate is insufficient for high-level planning. Decision-makers lack the contextual "why" such as funding bottlenecks or environmental clearances linked directly to the map location.
- Data Sovereignty & Security: Government bodies deal with sensitive data that cannot be offloaded to public cloud APIs (like OpenAI or Google Maps). There is a critical gap for secure, air-gapped, and offline-first AI solutions.
- Fragmented Tooling: Currently, no single open-source framework seamlessly integrates Optical Character Recognition (OCR), NLP, and GIS visualisation specifically tailored for the Indian public sector.

IV. Methodology

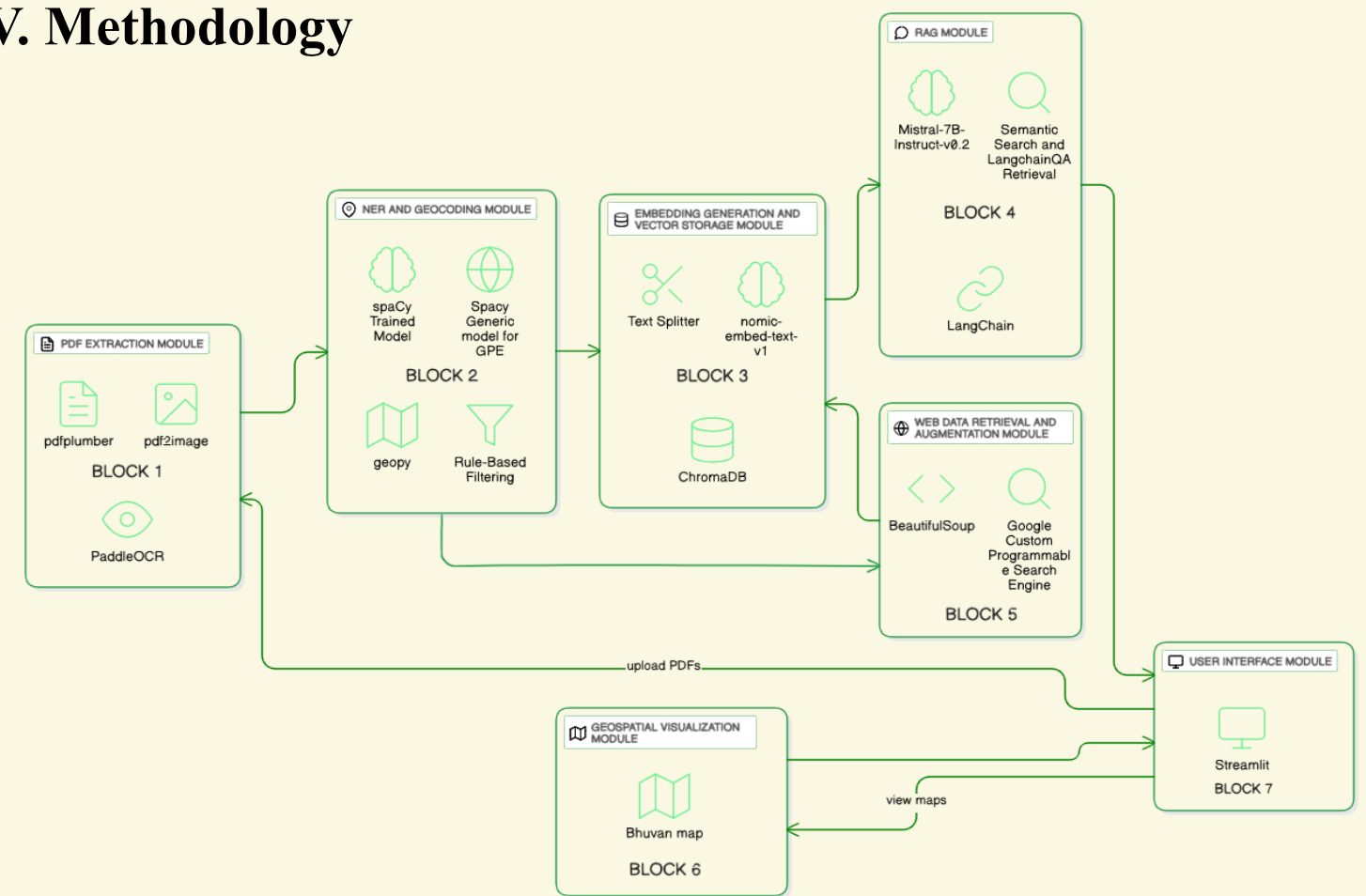


Fig. 1. Architecture of the framework.

- Block 1: Ingestion: Hybrid pipeline using pdfplumber (digital text/tables) and PaddleOCR (scanned images) to digitize diverse government reports.
- Block 2: Spatial Intelligence: Custom spaCy NER model fine-tuned on Indian Census data. Uses GeoPy and hierarchical validation (Town in District) to prevent false positives.
- Block 3: Vector Storage: Splits text into semantic chunks; encodes them using nomic-embed-text-v1 and stores them in ChromaDB.
- Block 4: RAG Engine: A local, air-gapped Mistral-7B LLM retrieves relevant vector chunks to generate context-aware summaries ("Urgent Issues") offline.
- Block 5: Web Augmentation (Optional): Integrates Google Custom Search to verify local data against trusted online sources (if connectivity permits).
- Block 6: Visualization: Maps extracted entities onto ISRO Bhuvan WMS (Satellite/Base layers) using OpenLayers for interactive analysis.
- Block 7: User Interface: A low-code Streamlit frontend enabling non-technical users to upload PDFs and view insights without GIS training.

V. Results and Analyses

Performance Metrics:

- Validation F1: 0.906 (High accuracy on training data).
- Real-World Precision: 1.000 (Zero false positives; system prioritizes mapping accuracy over quantity).

Geocoding Accuracy: Achieved >95% success by using hierarchical validation checks (e.g., confirming Town belongs to the correct District).

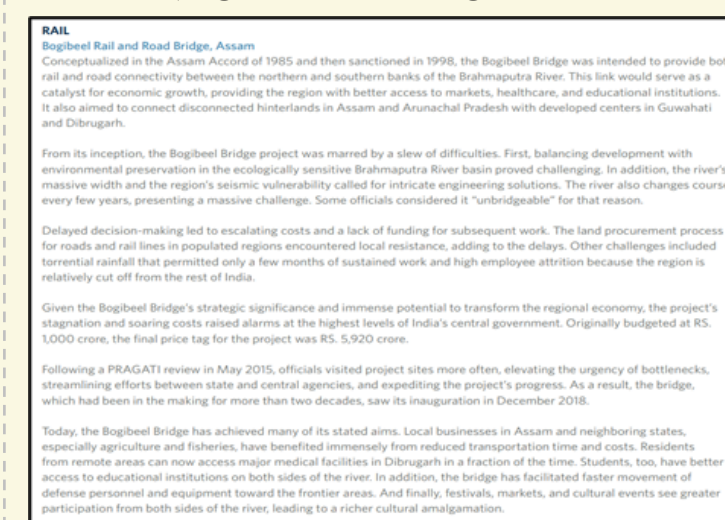


Fig. 2. Sample snapshot of a pdf document.

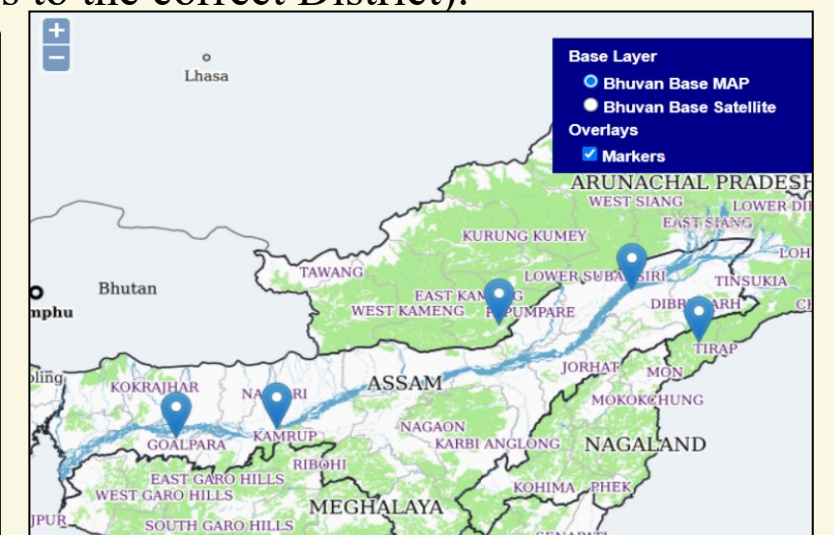


Fig. 3. Extracted entities visualized on ISRO Bhuvan WMS Base Layer.

Case Study (Bogibeel Bridge):

- Spatial Output: Accurately mapped key locations: Dibrugarh, Guwahati, and Brahmaputra River.
- RAG Insights: Automatically extracted "Urgent Issues" such as land procurement delays and cost escalation (from ₹1,000 Cr to ₹5,920 Cr).

Efficiency: The entire pipeline runs seamlessly on a standard mid-range laptop (16GB RAM), proving viability for offline field deployment.

VI. Conclusion & Future Scope

- Conclusion:** Developed a privacy-focused, offline framework that democratizes geospatial analysis with 1.000 Precision. The solution is lightweight, open-source, and easily deployable on standard hardware.
- Future Scope:** Extending capabilities to extract complex geometries (roads/polygons) and enabling direct GIS interoperability via GeoJSON/Shapefile exports.

References

- [1] R. S. Purves et al., "Geographic information retrieval: progress and challenges in spatial search of text," Foundations and Trends in Information Retrieval, 2018.
- [2] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," arXiv:2005.11401, 2020.

Acknowledgement: I sincerely express my gratitude to the National Remote Sensing Centre (NRSC), ISRO, Hyderabad, for providing the internship opportunity, essential infrastructure, and technical guidance required to successfully complete this research work.